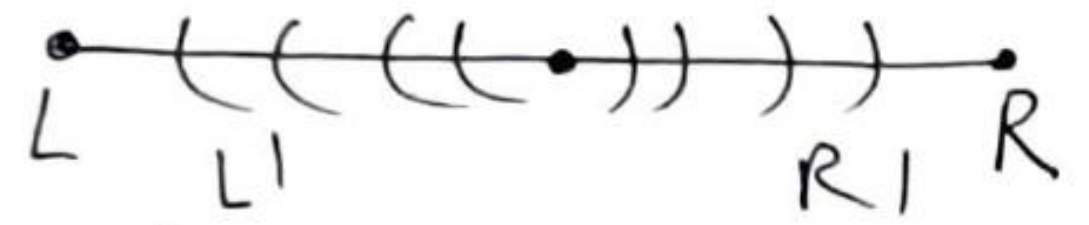


Golden Section Search Method

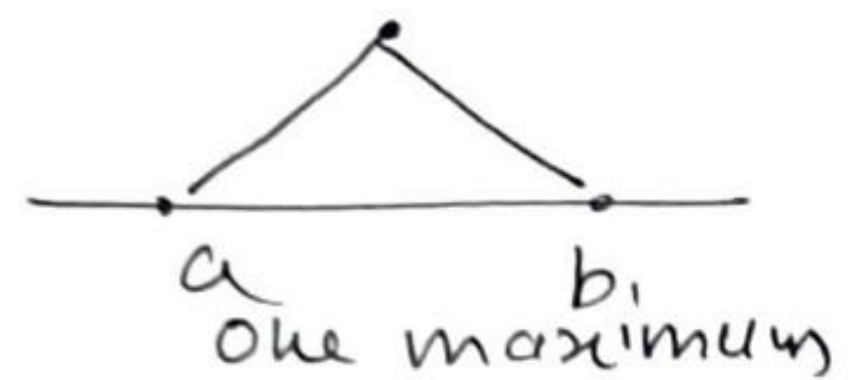
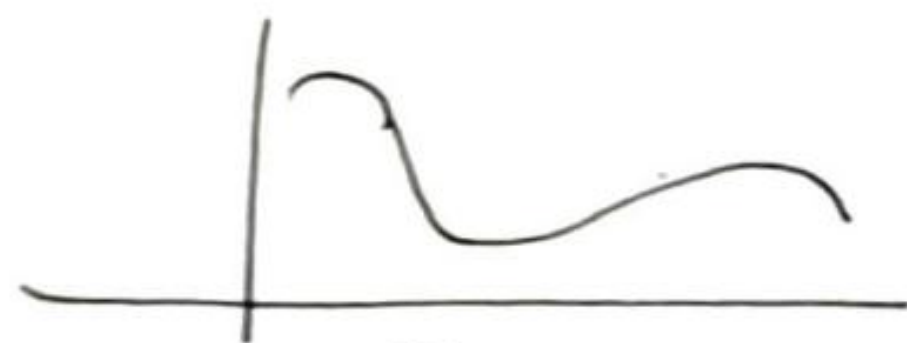
(Unconstrained optimization) ①

The golden-section search is an efficient way to progressively reduce the interval locating the minimum/maximum.

Golden Section is a technique to find out the extremum (maximum or minimum) of a strictly unimodal function.



Unimodal function only min/max in particular interval $[a, b]$



one max or one minimum

The Golden Ratio (also known as the Golden Section, Golden Mean, Divine Proportion or Greek letter Φ, ϕ) exists when a line is divided into two parts and the longer part (a) divided by the smaller part (b) is equal to the sum of (a) + (b) divided by (a), which both equal 1.618.

$$\frac{a}{b} = \frac{a+b}{a}$$

Difference between Golden Section method & Fibonacci method:-

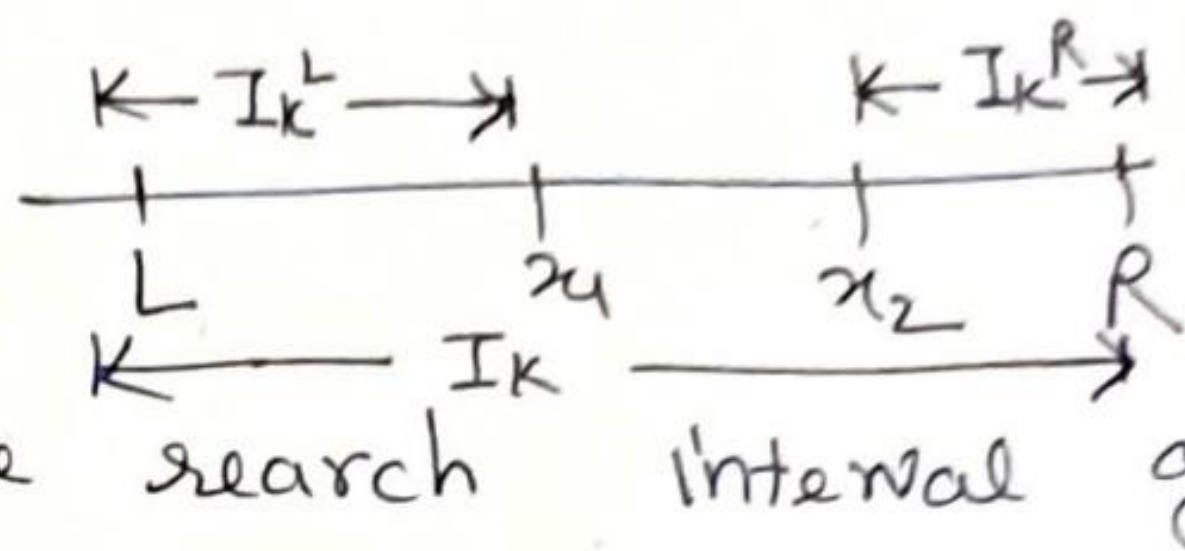
In Fibonacci method, we have
 → the ratio for the reduction of interval is $\frac{F_{n-1}}{F_n}$ constant.
 → the number of subintervals (iterations) ^(n = ?) is predetermined and based on the specified tolerance.

While in Golden Section method:-

→ the ratio of interval is constant.
 → Number of iterations are infinite, of course you may fix $n = 6, 7$ in advance.

Notations:-

$I_k = R - L$ length of the search interval of k^{th} iteration.

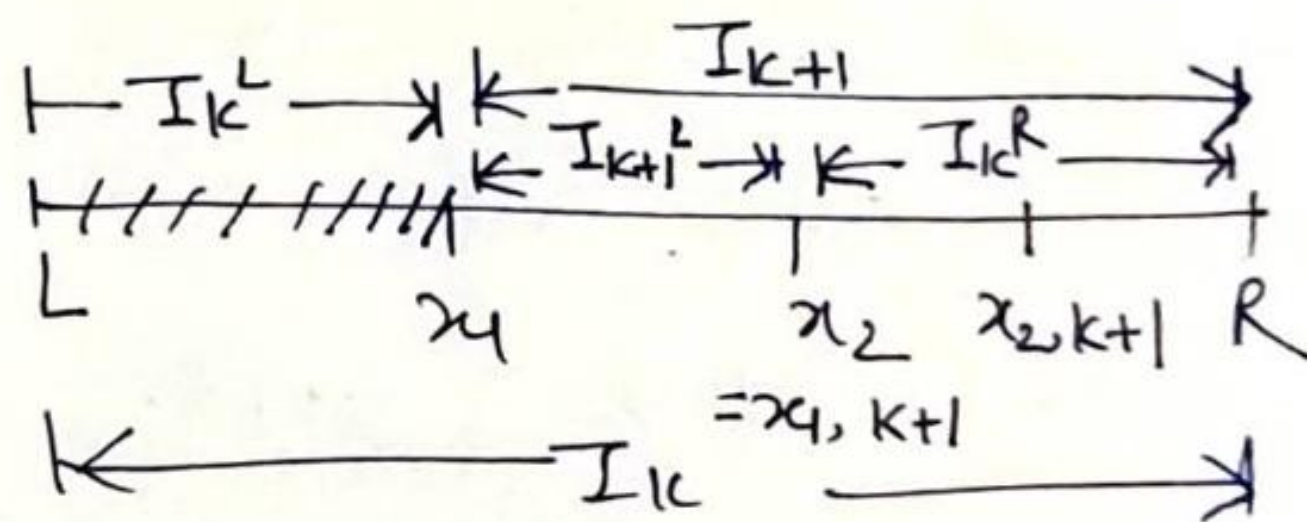


$I_k^L = x_{1,k} - L$ length of the left part of the search interval I_k .

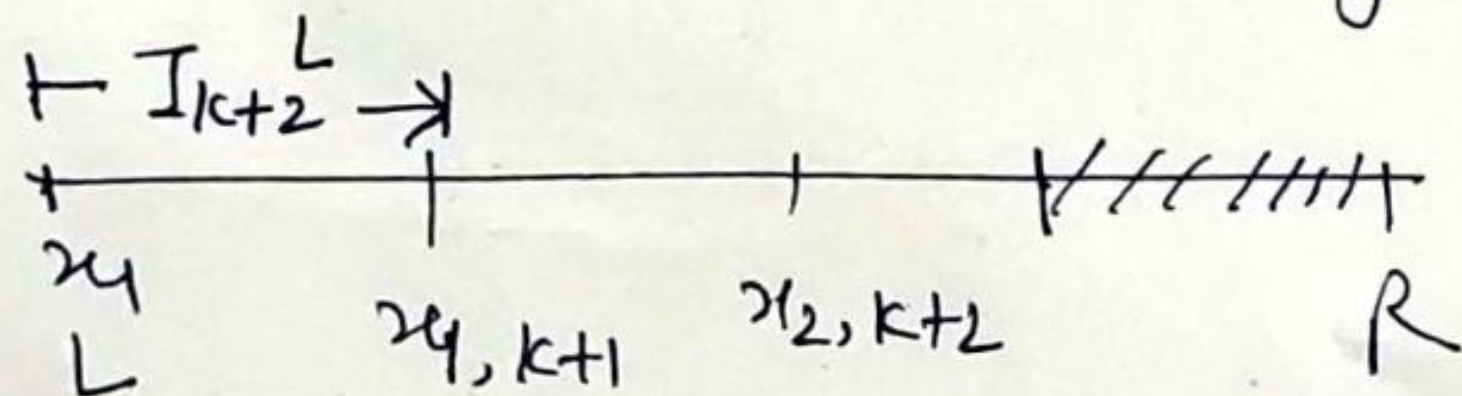
$I_k^R = R - x_{2,k}$ length of the right part of the search interval I_k .

Suppose x_1 moves towards right

at $(k+1)^{th}$ iteration



Suppose x_2 i.e. $x_{2,k+1}$ moves left



How to choose x_1 and x_2
choose two points x_1 and x_2 in between
the L and R according to the following rule:-

1) $I_k^L = I_k^R$, for all k .

2) Only point either x_1 or x_2 is computed
for instance,

$x_{1,k+1} \rightarrow x_{2,k}$ and $x_{2,k+1}$ is computed for I_{k+1}^R

$x_{2,k+1} \rightarrow x_{1,k}$ and $x_{1,k+1}$ is computed for I_{k+1}^L

such that $I_k = I_{k+1} + I_{k+2}$, for all k .

3)
$$\frac{I_k}{I_{k+1}} = \frac{I_{k+1}}{I_{k+2}} = C \text{ (constant) for all } k$$

This is called golden section criterion.

Since $I_k = I_{k+1} + I_{k+2}$

$$\Rightarrow \frac{I_{k+1}}{I_{k+2}} = \frac{I_{k+1}}{I_{k+2}} + 1$$

$$\Rightarrow \left(\frac{I_k}{I_{k+1}} \right) \left(\frac{I_{k+1}}{I_{k+2}} \right) = C + 1$$

$$C^2 = C + 1$$

$$\Rightarrow C^2 - C - 1 = 0$$

$$C = \frac{1 \pm \sqrt{5}}{2}$$

$$C = 1.618$$

Since C can not be negative.

we get $C = 1.618$

Remark:-

This number $c = 1.618$ is called golden section ratio.

$$x_1 + x_2 = L + R$$



Consider an initial search be $[L, R]$ then

$$\frac{R-L}{x_2-L} = \frac{x_2-L}{x_2-x_1} = 1.618$$

$$\Rightarrow \frac{R-L}{x_2-L} = 1.618$$

$$\Rightarrow x_2 = L + 0.618(R-L)$$

$$x_1 + x_2 = L + R$$

Golden Section Rule algorithm

For minimization

Let initial interval $[a, b]$ be given and the number of iteration be n or tolerance error ϵ .

- 1) Set $L = a$; $R = b$; $k = 1$ (iteration number)
- 2) Compute x_1 and x_2 as

$$x_2 = L + 0.618(R-L) \quad \text{and} \quad x_1 = L + R - x_2$$

If $f(x_1) < f(x_2)$ then

$$L_{k+1} = L_k;$$

$$R_k \leftarrow x_2; \quad x_2 \leftarrow x_1$$

i.e. preserve L and
Update R with x_2

If $f(x_2) \leq f(x_1)$ then

$$R_{k+1} = R_k$$

$$L_{k+1} \leftarrow x_1; \quad x_1 \leftarrow x_2$$

i.e. Preserve R and
update L with x_1

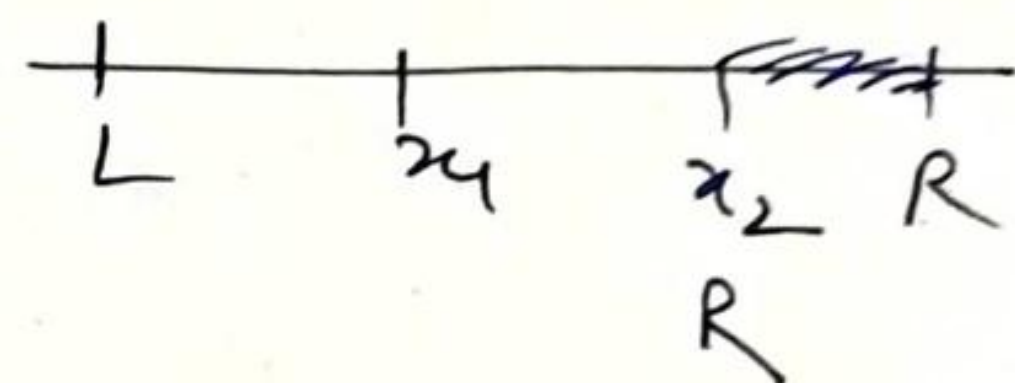
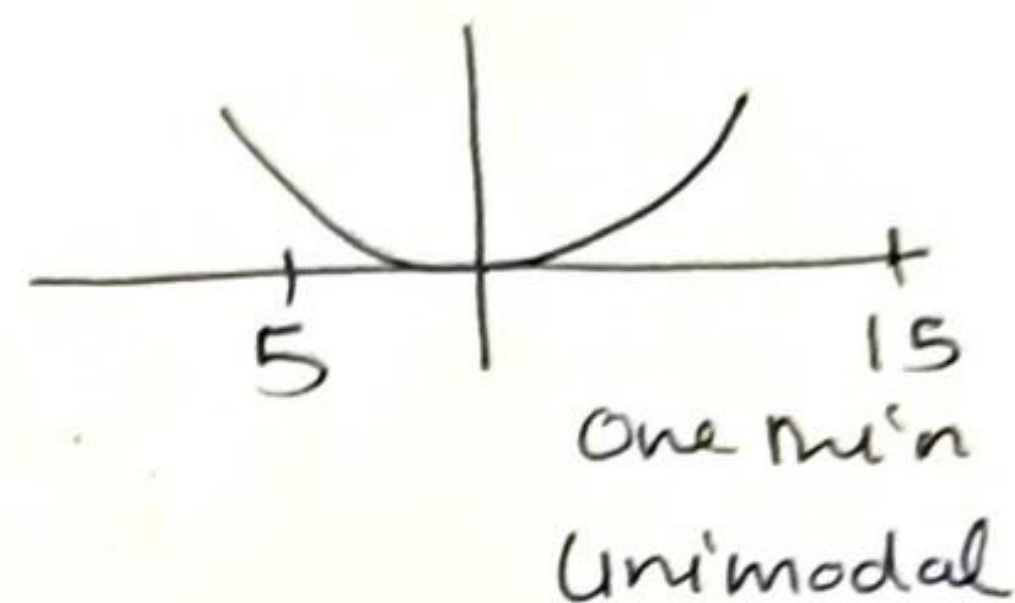
Repeat Step 2 until requirement meet.

③

Example:-

Minimize $f(x) = x^2$ over $[-5, 15]$ using Golden Section method. Take $n=7$.

k	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L/R
1	-5	15	2.64	7.36	6.9696	54.17	L
2	-5	7.36	-0.28	2.64	0.0784	6.9696	L
3	-5	2.64	-2.08	-0.28	4.3264	0.0784	R
4	-2.08	2.64	-0.28	0.84	0.0784	0.7056	L
5	-2.08	0.84	-0.96	-0.28	0.9216	0.0784	R
6	-0.96	0.84	-0.28	0.16	0.0784	0.0256	R
7	-0.28	0.84	0.16	0.4	0.0256	0.16	



$$\begin{aligned} x_2 &= L + 0.618(R-L) \\ x_1 &= L + R - x_2 \end{aligned}$$

Thus $x_{\min} \in [-0.28, 0.84]$

$$\text{Hence } x^* = \frac{-0.28 + 0.84}{2} = 0.28$$

$$f(x^*) = 0.00784$$

Ex:- Min $f(x) = x(x-1.5)$ in $[0, 1]$ by Golden Section rule with interval of uncertainty as 0.3.

Sol!

Given

that $\epsilon = 0.3$

$\epsilon = R - L \neq 0.3$

k	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L/R
1	0	1	0.382	0.618	-0.42708	-0.54508	R
2	0.382	1	0.618	0.764	-0.54508	-0.5623	R
3	0.618	1	0.764	0.854	-0.5623	-0.55168	L
4	0.618	0.854	0.708	0.764	-0.56074	-0.5623	

$L_4 = 0.854 - 0.618 = 0.236 < \epsilon = 0.3$
So we stop.

$I_k = I_{k+1} + I_{k+2}$

Thus,

$x_{\min} \in [0.618, 0.854]$ and

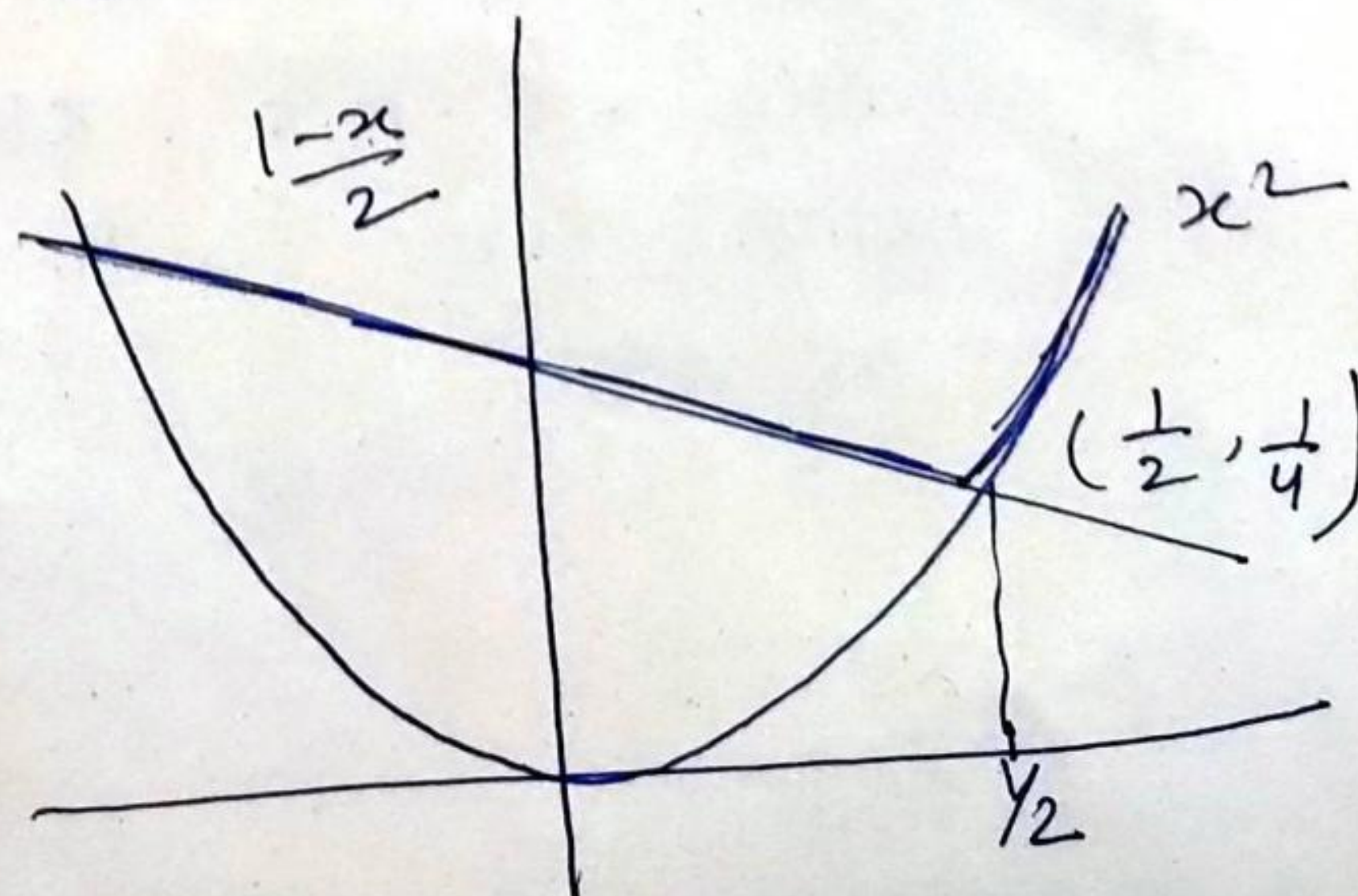
$E_{\min} \leq \min(-0.56074, -0.5623)$
 $= -0.5623$

$x^*_{\min} = \frac{0.618 + 0.854}{2} = 0.736$

$f(x^*) = 0.1943$

Q03:- Perform 6 iterations of the Golden Section method to minimize $f(x) = \max\{x^2, \frac{1-x}{2}\}$ over $[-1, 1]$ and hence identify the interval final interval I_6 .

$f(x) = \left\{ x^2, \frac{1-x}{2} \right\}$



Q. 10 min $f(x) = \begin{cases} \frac{1-x}{2}, & x < \frac{1}{2} \\ x^2, & x \geq \frac{1}{2} \end{cases}$ over $[-1, 1]$ (9)

K	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L/R
1	-1	1	-0.236	0.236	0.618	0.382	R
2	-0.236	1	0.236	0.528	0.382	0.27878	R
3	0.236	1	0.528	0.708	0.27878	0.50126	L
4	0.236	0.708	0.416	0.528	0.292	0.27878	R
5	0.416	0.708	0.528	0.596	0.27878	0.35522	L
6	0.416	0.596	0.484	0.528	0.258	0.27878	

Thus $x_{min} \in [0.416, 0.596]$

Hence, $x^* = \frac{0.416 + 0.596}{2} = 0.506$

$\min f(x^*) = 0.2560$

Qo:- Minimize $f(x) = x^2 - 2.6x + 2$; $x \in [-2, 3]$
Using Golden Section method. Perform 6 iterations.

Sol:-

k	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	WR
1	-2	3	-0.09	1.09	2.2421	0.3541	R
2	-0.09	3	1.09	1.82	0.3541	0.5804	L
3	-0.09	1.82	0.64	1.09	0.7456	0.3541	R
4	0.64	1.82	1.09	1.37	0.3541	0.3149	R
5	1.09	1.82	1.37	1.54	0.3149	0.3676	L
6	1.09	1.54	1.26	1.37	0.3116	0.3149	

Thus $x_{\min} \in [1.09, 1.54]$

Hence, $x^* = 1.31$ &

$$\min f(x^*) = 0.31490$$